CS 59300
Application of Deep Learning
Course Projects

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Three Types

- Application based
  - Apply the pre-built DL model(s) or the existing API(s)
  - Focus on the novelty and the usefulness
- Model based
  - Develop/train a new model or fine-tune the pre-trained model
  - Focus on data collection and performance evaluation
- Research based
  - Discover something new in the research
  - Focus on re-implementing a research paper and advancing the knowledge

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Basic Requirements

- Build an application or develop the research that applies the technology of deep learning
- Examples
  - Real-time speech translation
  - Digital assistants (similar to Google Assistant and Amazon Alexa) for students in a course
  - Autonomous driving
  - Superhuman Go playing
  - Applications based on ChatGPT
  - Defense against adversarial attacks in speech recognition
  - …

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What are My Expectations for Application-Based Course Group Projects?

- Useful or fun
  - Other people would like to use it
- Very easy to apply or use
  - For example, Web-based application
- Well-design and well-coded
  - Apply deep learning in a reasonable way
  - Follow code standard and code is easy to read
- Team-work
  - Evenly divide the work

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What are My Expectations for Model-Based Course Group Projects?

- Useful or fun
  - Other people would like to use it
- Data collection and pre-processing
- Model selection and training
  - How to determine hyper-parameters of a model
- Performance evaluation
  - For example, F1 score for binary classification
- Well-design and well-coded
- Team-work

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What are My Expectations for Research-Based Course Group Projects?

- Interesting research problem
  - Other researchers are interested
- Understand what has been done in the field
  - Read relevant research papers
- If the problem is not totally new, most likely it needs to re-implement the work in a paper
  - Find the paper with the code (https://paperswithcode.com/)
  - Compare your work with the existing work
- Well-design and well-coded, and team-work

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Evenly Divide the Work

- Based on GitHub **pull requests**
- If you commit no code or few code to the project, you will lose most of points for project review and final demo
- My recorded video "GitHub Pull Request and Merge Conflict Video (19:47)" under "Course Project" in Brightspace
- (YouTube) "**Pull Requests in VS Code**" by Visual Studio Code
- https://www.youtube.com/watch?v=LdSwWxVzUpo&ab_channel=VisualStudioCode

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Important Notes

- It is ok to start the project with an existing open source project.
  - I would expect that you will apply the existing deep learning technologies or project(s)
- But you will need to write the additional code, or build the hardware in the project, or connect multiple projects into one!
- Simply copying an existing open source project and running it is NOT acceptable!
- Not use the same project in two different courses

Some Considerations

- Publish the application at Hugging Face Spaces or Google Cloud
  - https://huggingface.co/spaces
- Present your project at Data Science Week
  - https://sites.google.com/view/data-science-week-2024
- Present your project at PFW Student Research and Creative Endeavor Symposium next year
  - https://www.pfw.edu/centers/irsc/research-symposium/presenters

Student Posters From This Course in Previous Semesters

- S. Bommireddy, A. Mishra, Z. Chen, and A. Coronado, "Donna: The Virtual Assistant of PFW," poster presentation at PFW Student Research and Creative Endeavor Symposium, March 2025. **Won the Student DEI Research Award**.
- N. Rajappa, M. Vivas, D. Lybarger, and Z. Chen, "Learning Assistant for Diverse Learning Needs," poster presentation at PFW Student Research and Creative Endeavor Symposium, March 2025. **Won the Student DEI Research Award**.
- P. Dayananda, A. Nagaraj, J. Bollineni, S. Kuna, N. Talastha, and Z. Chen, "DevAssistAI: An AI Pair Programmer," poster presentation at PFW Student Research and Creative Endeavor Symposium, March 2025.
- N. Singh, D. Parveez, S. Sharma, K. Mallick, and Z. Chen, "PFW CS Chatbot," poster presentation at PFW Student Research and Creative Endeavor Symposium, March 2024. **Won the Third Place for Graduate Student Awards**.
- Y. Bhandare, W. Har, M. Trovinger, A. Atre, and Z. Chen, "Over the Air Voice Translation," poster presentation at PFW Data Science Week, December 2022. **Won the Third Place in the Poster & Short Talks competition**.

Please Follow Course Project Rubrics

Q & A